EDUCATION

CARNEGIE MELLON UNIVERSITY PITTSBURGH, PA
Doctor of Philosophy - Robotics, School of Computer Science Expected 05/2026
Master of Science - Robotics, School of Computer Science 05/2025
Specializations: Human-Robot Interaction (HRI), Design Research, Inclusive Design.

STANFORD UNIVERSITY STANFORD, CA
Bachelor of Science - Mechanical Engineering, School of Engineering 06/2021
Specializations: Dynamic Systems & Controls.

SKILLS

User Research | Experimental Studies. Interview, Focus Groups, Diary Study, Survey (Qualtrics, Prolific), Usability (Contextual Inquiry, A/B Testing, Heuristic Evaluation). Participatory Design. Data Analysis: R, JMP, Matlab, QDA (ATLAS.ti).

Design & Prototyping | Wireframing & Journey Mapping (Figma, Adobe Xd, Miro). CAD/CAE: Solidworks, Fusion 360. Fabrication (3D Printing, Laser Cutting). Unity 3D, Processing.

Robotics & AI | Python. Agent Training (Reinforcement Learning from Human Feedback (RLHF), Imitation Learning. Frameworks & APIs (PyTorch, TensorFlow, ROS, Gymnasium). Physics Simulation (MuJoCo). Robots (Pioneer 3DX, Create 2, Misty II).

Software & Firmware | Java, JavaScript (React.js, Node.js), REST, Fetch. Embedded C/C++. Microcontrollers (STM32, Arduino), Circuit Design.

RELEVANT WORK EXPERIENCE

CARNEGIE MELLON UNIVERSITY, ROBOTICS INSTITUTE PITTSBURGH, PA
Human-Robot Interaction Researcher (Safety-Critical & Inclusive HRI) 07/2021 - Present

- Spearheaded mixed-methods UX research & end-to-end design for embodied & agentic AI across autonomous navigation, last-mile delivery, & personal well-being with 100+ diverse users.
- Architected multimodal sensing pipelines (computer vision; wearable: Garmin's *Monkey C*) & decision-making approaches (LLM-driven: *OpenAI API* & Preference Learning: *Bayesian Contextual Bandits*); evaluated usability across 15+ end-user deployments focused on real-time, personalized robot behaviors on personal well-being [Github Repo].
- Co-investigated guide robot path planning strategies on blind and low-vision users; developed scalable design guidelines for usability, robot behavioral paradigms (i.e., legible motion, shared autonomy, & communication) to preserve user safety & agency.
- Executed large-scale (n=70) comparative preference studies to evaluate impacts of agent behavior on user trust; established UX benchmarks for collaborative workflows of human-agent teams.
- Developed production-ready React.js interface integrated with Preference Learning to adjust model behavior based on real-time user engagement; optimized social robot behaviors for personal well-being settings [Github Repo].
- Optimized research operations by increasing diary study compliance from estimated 60% to over 90% rigorous user study design and iteration, thereby ensuring high-quality longitudinal data.
- Engineered low-latency motor control strategies for human-safe haptic wearables; optimized for haptic transparency in VR experiences.

Course Assistant & Instructor, Sensing & Sensors (Grad-Level), HRI (UGrad-Level) 07/2023 - 05/2025

- Directed & advised research operations for 50+ cross-functional teams (Engineering, Design, Psychology) across 19 distinct projects, providing oversight on methodology, experimental design, & data analysis.

- Developed & delivered advanced lecture on **sensor fusion** & **state estimation** (Odometry & Kalman Filtering) to 20+ graduate students, addressing gaps in technical understanding.

SHAPE LAB, STANFORD UNIVERSITY — STANFORD, CA
HCI & HRI Undergraduate Researcher (Accessible Haptics) — 06/2019 - 06/2021

- Engineered **novel robot-mediated haptic devices** & architected inclusive, **multimodal interaction paradigms** to enable users with visual disabilities to interact with complex digital information.
- Identified interaction patterns & co-designed interaction strategies for wearable haptics; co-authored two publications.

AWARDS & HONORS

1. **Best Paper Honorable Mention Award** (2026): Top 5% at ACM/IEEE HRI Design Track.
2. **HRI Pioneer** (2026): Recognized as 1 of 22 elite global HRI researchers for contributions to the future of Human-Robot Interaction.
3. **Cadence Future Innovators Scholarship** (2023).
4. **Uber Presidential PhD Fellow** (2022-2023).
5. Nomination for **Hoefer Prize for Writing in the Major** (2021), Mechanical Engineering, Stanford University.
6. **University of Washington College of Engineering Dean's Fellowship** (*declined*) (2021): Graduate fellowship award for top incoming doctoral applicants ($10k/yr) (Nominated by HCDE).
7. **University of Washington Top Scholar Fellowship** (**declined**) (2021): Award to attract outstanding students to the University of Washington ($2.5k).
8. **Stanford Mechanical Engineering Summer Undergraduate Research Institute** (SURI) (2019, 2020): Research fellowship ($7.5k) to conduct and present a ten-week, original research project.
9. **Top Achievers Award and Scholarship** (2017): Selective full university scholarship for stellar academic achievement, awarded by Botswana's Ministry of Education.
10. **American University Preparatory School Global Leadership Merit Scholarship** (2014-2017): Three-year academic scholarship for a student with exemplary leadership qualities.

PUBLICATIONS [*denotes equal contribution. **Google Scholar** for full papers.]

Book Chapters (Peer Reviewed)

1. Siu, A. F., Chase, E. D. Z., Kim, G. S. H., **Boadi-Agyemang, A**, Gonzalez, E. J., and Follmer, S. (2021). Haptic Guidance to Support Design Education and Collaboration for Blind and Visually Impaired People. In *Plattner H., Design Thinking Research. Understanding Innovation. Springer.*

Journal Articles (Peer Reviewed)

1. Shih, K.*, **Boadi-Agyemang, A.***, Carter, E.J., Steinfeld, A. (2025). A Multi-Method Investigation of Guide Robot Characteristics for Blind and Low Vision Users. In *ACM Transactions on Human-Robot Interaction*.

Conference Papers (Peer Reviewed)

1. **Boadi-Agyemang, A**., Moharana, S., Bennett, C. L., Carter, E. J., Carrington, P., & Steinfeld, A. (2026). Pluriversal Approach to Co-designing Delivery Robots with People with Disabilities. In *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. **Honorable Mention Award**.
2. **Boadi-Agyemang, A**., Schaldenbrand, P., Misra, V., Carter, E. J., Oh, J., & Steinfeld, A. (2025). If I Move, Do You Move? Investigating the Role of Interpersonal Synchrony in Human-Robot Joint Painting. In *2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*.
3. Han, V. Y., **Boadi-Agyemang, A**., Lin, Y., Lindlbauer, D., and Ion, A. (2023). Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices. In *The 35th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Short Papers (Peer Reviewed)

1. **Boadi-Agyemang, A.*** and Park, M.*. (2024). Designing Interactive Agents to Support Emotion Regulation in the Workplace through Guided Art-Making. In *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*.
2. **Boadi-Agyemang, A.**, Carter, E.J., Siu, A. F., Steinfeld, A., and Martinez, M.O. (2023). Understanding Experiences, Attitudes, and Perspectives towards Designing Interactive Creative Tools for Teachers of Visually Impaired Students. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.
3. Kim, L. H.*, **Boadi-Agyemang, A**.*, Siu, A. F., and Tang, J. (2020). When to Add Human Narration to Photo-Sharing Social Media. In *The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.
4. Chase, E. D. Z., Siu, A. F., **Boadi-Agyemang, A.**, Kim, G. S. H., Gonzalez, E. J., and Follmer, S. (2020). PantoGuide: A Haptic and Audio Guidance System to Support Tactile Graphics Exploration. In *The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*.
5. Chase, E. D. Z., Siu, A. F., **Boadi-Agyemang, A.**, Kim, G. S. H., Gonzalez, E. J., and Follmer, S. (2020). PantoGuide: A Haptic and Audio Guidance System to Support Tactile Graphics Exploration. In *The 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS).*

Workshop (Non-Archival, Lightly Reviewed)

1. **Boadi-Agyemang, A.***, Moharana, S.* (2025**). T**he Case for Em(body)ed Approaches to Disability-Centered Interaction Design. “Body Politics: Unpacking Tensions and Future Perspectives for Body-Centric Design Research in HCI” at *ACM Conference on Human Factors (CHI)*.
2. **Boadi-Agyemang, A**.*, Shih, K.* and Steinfeld, A. (2023). Discovering User Needs and Preferences for Guide Robots: Challenges and Preliminary Insights. In Workshop “Assistive Robots for Citizens” at *IEEE/RSJ Conference on Intelligent Robots and Systems (IROS*).

Theses

1. **Boadi-Agyemang, A**. (2024). Simulated Encounters of the Third Kind: Scenario-Based Approach to Designing Guide Robots. *Master’s Thesis. Carnegie Mellon University*.
2. Perez, G. A.*, **Boadi-Agyemang, A**.*, Mhatre, K.*, Wang, J.* (2021). Using a Vibrotactile Stimulation Device to Improve Knee Extension During Stance Phase in Children with Spastic Cerebral Palsy. *Undergraduate Capstone, Stanford University.* **Nominated for Hoefer Prize for Writing in the Major.**